Ternary Modality Contrastive Learning for Hyperspectral and LiDAR Data Classification

Shuxiang Xia, Xiaohua Zhang, *Member, IEEE*, Hongyun Meng, and Licheng Jiao, *Fellow, IEEE*

***Abstract*— In the domain of remote sensing image classification, single sensors are constrained by their sensing angles and information dimensions, rendering them incapable of fully capturing the intricate characteristics of ground objects. Different sensors can provide complementary information, significantly enhancing the performance of object classification. However, due to their unique physical observation principles and varying spatial–spectral resolutions, different modalities capture heterogeneous features of ground objects, resulting in a semantic gap issue when integrating modalities. This article designs a multimodal contrastive learning framework, starting with the preprocessing of hyperspectral image (HSI) to obtain its spatial and spectral modalities, and then combining these with the light detection and ranging (LiDAR) modality, forming a ternary modality contrastive learning framework that achieves deep semantic alignment between different modalities. Furthermore, to enhance the model's generalization ability, based on the neighborhood semantic similarity of HSI, we propose a spectral selection data augmentation method. Extensive experiments on four public datasets show that our method outperforms several other state-of-the-art (SOTA) methods in classification performance.**

***Index Terms*— Contrastive learning, data augmentation, hyperspectral image (HSI) classification, light detection and ranging(LiDAR), self-supervised learning.**

I. Introduction

HYPERSPECTRAL image (HSI), in contrast to multispectral imaging, is distinguished by its capability to provide a nearly continuous and high-resolution spectral information profile for each pixel. This enables the precise identification and analysis of subtle differences in materials. Furthermore, the aggregation of this pixel-wise spectral data facilitates the elucidation of the spatial distribution of various categories at the observation site. Classifying each pixel of HSI is a common task in the field of remote sensing [1], with applications in plant disease diagnosis [2] geological exploration [3], airborne surveillance [4], and so on [5], [6]. However, HSI classification faces the following issues.

1) HSI sensors provide detailed spectral information by capturing hundreds of narrow bands of the target's reflected light. Cloudy and foggy weather can scatter or block these light rays, leading to blurred images and loss of spectral information, which affects the accuracy of image classification.
2) HSIs reveal the spectral properties and chemical composition of ground objects, allowing for the precise identification and classification of different materials. However, the process of manual annotation can be subject to subjectivity, resulting in ground objects with the same chemical composition being assigned different labels, such as shrubs and forests. Moreover, the presence of analogous chemical constituents and spectral characteristics, as observed in entities like roads and houses, can exacerbate the complexity of classification tasks utilizing HSI. This similarity increases the challenge associated with discriminating among diverse ground objects.
3) HSIs, obtaining information solely from spatial and spectral dimensions, require a large number of samples to help models capture and understand the subtle differences that are difficult to discern with only a single data source.

To mitigate the challenges delineated previously, researchers have attempted to incorporate data from other sources, such as light detection and ranging (LiDAR) [7] and synthetic aperture radar (SAR) [8]. LiDAR, relying on laser light rather than sunlight, remains impervious to atmospheric obfuscations such as cloud cover or fog. It introduces additional 3-D structural information, which can improve the classification ambiguities arising from the subjective nature of manual annotations, particularly in instances involving ground objects with chemically similar compositions. The synthesis of LiDAR data with other information sources facilitates a multidimensional capture of the characteristics of ground objects, this integration empowers models to formulate more holistic representations of features, diminishing the dependency on extensive sample datasets. Therefore, an increasing number of researchers are exploring the joint classification of ground objects using a multimodal fusion of HSI and LiDAR data.

In conventional machine learning frameworks, feature extraction and selection processes are first performed

Manuscript received 2 April 2024; revised 28 May 2024; accepted 15 June 2024. Date of publication 20 June 2024; date of current version 9 July 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 61877066, in part by the Aero-Science Fund under Grant 20175181013, and in part by the Science and Technology Plan Project of Xi'an under Grant 21RGZN0010. *(Corresponding author: Xiaohua Zhang.)*

Shuxiang Xia, Xiaohua Zhang, and Licheng Jiao are with the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education, School of Artificial Intelligence, Xidian University, Xi'an 710126, China (e-mail: xh_zhang@mail.xidian.edu.cn).

Hongyun Meng is with the School of Mathematics and Statistics, Xidian University, Xi'an 710126, China (e-mail: menghy@xidian.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3417011

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:37:32 UTC from IEEE Xplore. Restrictions apply.

separately on the two different modalities of data (HSI and LiDAR). Then, features derived from both modalities are fused through mathematical models to form an integrated feature set. Finally, a classifier is used to perform ground object classification based on this integrated feature set. Yanshuang and Xiaoli [9] enhanced the precision of tree species classification within complex forest ecosystems through the application of independent component analysis (ICA) for feature extraction and the random forest algorithm for feature selection, culminating in the employment of a support vector machine (SVM) classifier for the final classification phase. Huang and Zhu [10] employed extended morphological profiles (EMPs) to extract spatial texture features from the two modalities, utilizing the random forest algorithm for the generation of decision trees, thereby achieving improved classification accuracies. Xu et al. [11] utilized morphological profiles to reduce the irregularities in LiDAR measurements. Rasti et al. [12] integrated features of two modalities into one framework using orthogonal total variation component analysis (OTVCA), promoting the smoothness of classification maps while maintaining spatial structure. Nonetheless, these methodologies for feature extraction and fusion are heavily contingent upon extensive domain expertise, necessitating manual adjustment of model parameters to accommodate the characteristic of disparate datasets—a process that is not only labor-intensive but also prone to the biases inherent in subjective human judgment. In addition, when multiple data sources are integrated to produce high-dimensional feature representations, it presents significant processing challenges for traditional machine learning algorithms [13], [14]. These algorithms typically exhibit their inherent limitations when dealing with data of high dimensionality and complexity.

Deep learning, with its end-to-end processing capability—directly learning from raw data to complete specific tasks, efficiently handling high-dimensional multimodal features, enhancing model generalization ability, and reducing the need for data preprocessing and manual model parameter tuning—has been widely applied in the field of hyperspectral and LiDAR data classification [15], [16]. For instance, Hang et al. [17] innovatively deployed coupled convolutional neural networks (CNNs) for the integration of HSI and LiDAR datasets, achieving a reduction in model parameters and enhancing the efficiency of feature fusion. Li et al. [18] proposed a three-stream CNN network to extract spatial, spectral, and elevation information from data, effectively integrating heterogeneous features in the kernel space using multisensor composite kernels. Fang et al. [19] unveiled a cross-modal spatial–spectral enhancement module aimed at amplifying the attributes of features derived from varied data sources. Feng et al. [20] conceived a dynamic scale feature extraction technique that dynamically identifies and selects the most appropriate scale features for input. Recognizing the inherent challenges posed by CNNs in capturing global feature interactions due to their localized focus, Zhao et al. [21] integrated transformers to enhance the capture of long-range dependencies. Transformers, through their self-attention mechanisms, excel in identifying correlations across any two positions within the input sequence, irrespective of the specific nature of the data, thus enabling a more nuanced exploration of the interdependencies between the two modalities. In a similar vein, Wang et al. [22] proposed a novel modal attention module that fully considers the relevance and heterogeneity between different data modalities, achieving complementary advantages between HSI and LiDAR data. In addition to the aforementioned strategies, dynamic group convolution [23], Gabor filters [24], and graph convolutions [25] have also demonstrated excellent classification results in the field of joint classification of HSI and LiDAR data. However, these methodologies predominantly subscribe to supervised learning paradigms, exhibiting an overt dependence on annotated training datasets and necessitating a voluminous and diverse corpus of samples to ensure robust model generalizability. In addition, these methods typically use attention mechanisms to bridge the semantic gap between modalities, concentrating on harmonizing the feature similarities across diverse modalities by dynamically allocating weights to key information. The association between modalities often depends on the same class labels in both modalities, making it difficult for attention mechanisms to explore deeper relationships between the two modalities.

Self-supervised learning can utilize the inherent data features of unlabeled data to automatically generate supervisory signals, and then use these pseudo-labels to train models to mine the structure and information within the data, enabling the model to capture the generic features of the data. In application to specific tasks, this paradigm necessitates only a minimal set of manually annotated samples for model refinement [26], [27], substantially mitigating the challenges posed by dataset sample scarcity. In the domain of joint classification of HSI and LiDAR data, research on self-supervised learning is still in its preliminary exploration stage. These methods adopt the strategy of inputting HSI as image blocks into the model, primarily to utilize the spatial information of the HSI modality. However, when the input image blocks contain pixels belonging to different classes from the target pixel, this method is prone to introducing features of nontarget categories into the feature representation of the target pixel, thereby negatively affecting the classification accuracy. Moreover, HSI determines the spatial information of each pixel by capturing the spectral reflectance of the target object across multiple bands through imaging equipment. However, due to the inherent phenomenon of "same spectrum, different objects" in HSI—where objects of different categories may have similar spectral features at certain specific bands—these specific bands can also interfere with the spatial information. Finally, regarding data augmentation techniques, current popular methods such as rotation and cutout mainly operate on the spatial dimension of images to increase the diversity of training data. While these methods have improved the model's robustness to spatial variations to a certain extent, they have neglected the richness of spectral information in HSI. The exploration into spectral-dimension data augmentation techniques remains limited, curtailing the model's capacity for adaptation and learning from spectral variations, thus failing to fully exploit the latent potential of HSI.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:37:32 UTC from IEEE Xplore. Restrictions apply.

In addressing the challenges identified previously, this research introduces a novel ternary modal contrastive learning framework that incorporates an additional spectral modality. This innovative approach is designed to effectively mitigate the negative effects of heterogeneous pixels on the classification accuracy of target pixels. Unlike conventional methods that utilize image blocks as inputs, our framework adopts a unique strategy by randomly selecting a single spectral band during each iteration. This method strategically minimizes the bias often encountered in spatial information interpretation, which arises from the phenomenon where different objects exhibit similar spectral characteristics at specific bands. In addition, this technique enhances sample diversity through the introduction of randomness in band selection throughout the training process. Moreover, previous methods have consistently employed traditional data augmentation techniques for RGB images to enhance the spatial diversity of samples. Considering the high semantic similarity of spectral features between adjacent pixels in HSI, we have designed specialized data augmentation strategies to enhance the spectral diversity of HSI. This strategy is crafted to harness the dense spectral information inherent in HSI through a direct yet efficacious approach, thereby uncovering the common features of adjacent spectra.

1) We propose a ternary modal contrastive learning framework that, by integrating a spectral modality, significantly enhances the representational capability of target pixels within the feature space and effectively reduces the interference effect of heterogeneous pixels during the classification process.
2) We have developed an innovative data augmentation approach according to the semantic similarity among adjacent pixels within HSI. This method, through enriching the diversity of the spectral dimension, substantially augments the generalization capacity of the model.
3) We conducted experiments on four datasets combining HSI and LiDAR data, and the results demonstrate that our approach outperforms other state-of-the-art (SOTA) methods, particularly on the Houston and Augsburg datasets.

The rest of this article is organized as follows. Section II shows the related work on contrastive learning and cross-modality contrastive learning. Section III shows the details of our model. The experimental description and the result analysis on four public datasets are provided in Section IV. Finally, Section V summarizes the conclusion of this article.

II. Related Work

A. Self-Supervised Contrastive Learning

In the field of self-supervised learning, an effective paradigm involves generating pseudo-labels by leveraging the semantic information inherent in the data itself to train models, which can then be fine-tuned with only a minimum of manually annotated samples for specific tasks. This sphere is principally bifurcated into two predominant categories: predictive and contrastive self-supervised learning. Predictive self-supervised learning endeavors to enable models to anticipate the inherent attributes of input data. This includes, but is not limited to, the rotation angle of an image [28], the content of areas obscured by masks in an image [29], and the content of absent frames within a video [30]. Methods in this category design specific loss functions based on the needs of particular tasks to encourage models to learn features closely related to those tasks. Nonetheless, an inherent limitation of this methodology is its potential for generating features of high specificity, which may hamper the model's performance accuracy when transitioning from an upstream task to a downstream task of diminished relevance.

Conversely, contrastive learning delineates a methodology that discriminates between diverse samples by diminishing the distance between pairs of positive samples while augmenting the distance between negative samples. This compels the samples to capture features persistently manifest across varied conditions, disregarding those that emerge solely under specific circumstances. This modality predominantly concentrates on the data itself, thereby garnering generalized features of the data. In the domain of HSI classification [31], [32], Hou et al. [33] pioneered the application of contrastive learning, leveraging a substantial corpus of unlabeled samples to mitigate the challenge of insufficient labeled data in this area. Zhang et al. [34] employed residual 3-D convolution as the feature network to bolster the model's proficiency in extracting spatial–spectral information. Furthermore, Huang et al. [35] designed an adaptive 3-D swim transformer based on the Swim transformer and the characteristic attributes of HSI and utilized a hierarchical contrastive learning method to fully exploit the multiscale semantic information of images. Most of the aforementioned methods help in mining data features by utilizing networks with stronger feature extraction capabilities, ignoring the spatial–spectral diversity inherent in HSI. Guan and Lam [36] designed a cross-domain contrastive learning (XDCL) framework to mine shared features effectively for class recognition between the spatial and spectral domains, which significantly improved experimental accuracy in a large number of experiments.

B. Cross-Modality Contrastive Learning

With the ascension in computational power of graphics processing units and the swift progression of large models, contrastive learning has gained increasing attention in the field of multimodal learning due to its significant advantages in augmenting feature representation capabilities, fostering interactions across modalities, effectively utilizing unlabeled data, and its high flexibility and universality. Radford et al. [37] elucidated the model's robust generalizability across diverse visual tasks via pretraining through contrastive learning on an expansive dataset of image–text pairs, obviating the requirement for task-specific fine-tuning. Li et al. [38] introduced momentum distillation to address the inevitable presence of noise within large-scale image–text datasets, crafting a more efficacious training framework. Lu et al. [39] devised co-attentional transformer layers to enable effective interactions between two modalities at different network levels, capturing

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:37:32 UTC from IEEE Xplore. Restrictions apply.

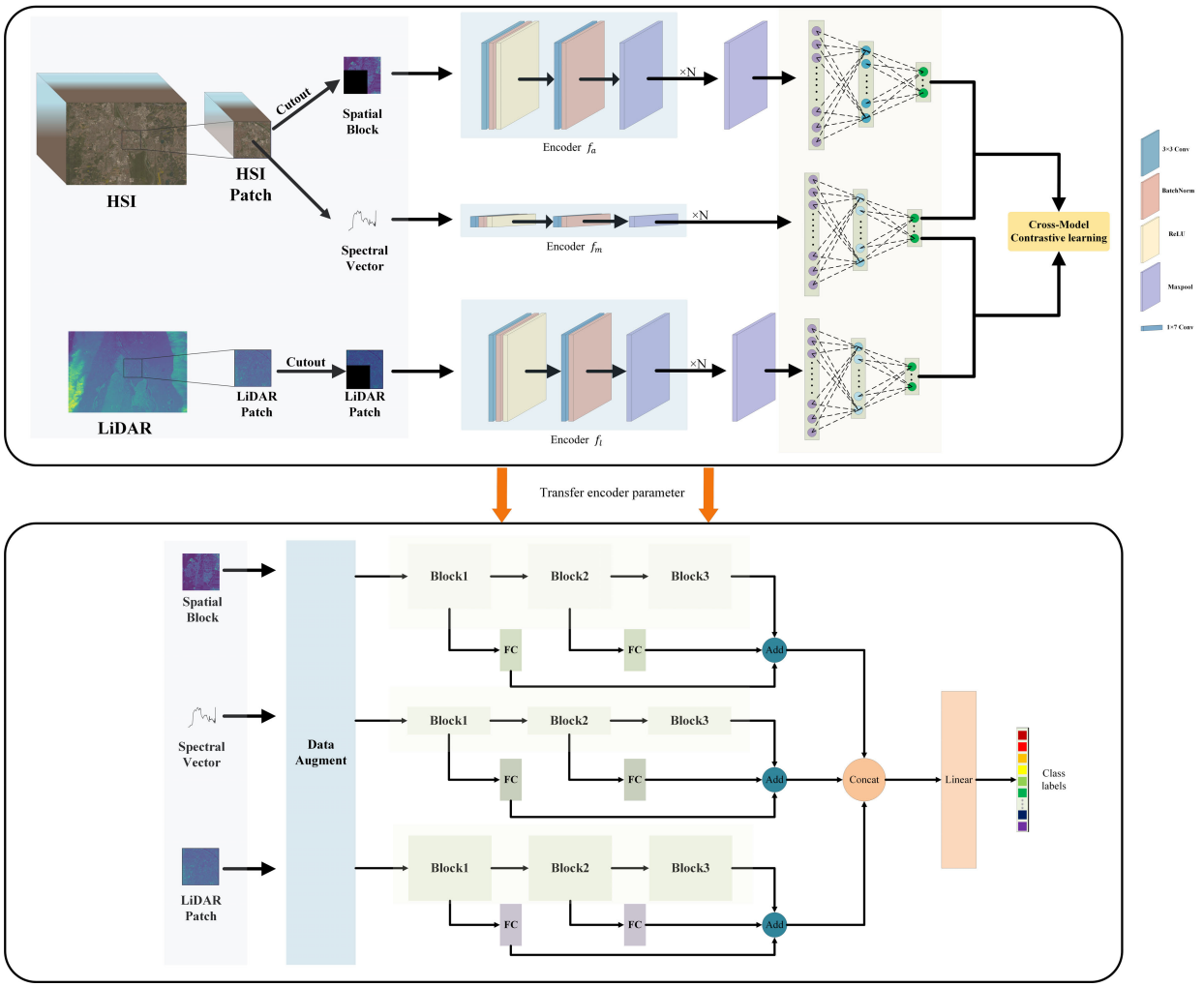


Fig. 1. Whole framework of the model for HSI and LiDAR classification.

richer cross-modal semantic information. Bao et al. [40] introduced the multiway transformer to bolster the capability of processing different modal inputs, culminating in superior performance across a multitude of multimodal tasks. Some researchers have begun to explore the introduction of contrastive learning into the field of multimodal remote sensing classification. Hang et al. [41] innovatively applied contrastive learning to analyze HSIs and LiDAR data from identical geographical locations as disparate perspectives of the same ground object, yielding remarkable outcomes with a scant number of labeled samples. Wang et al. [42] developed a bilinear attention fusion to capture second-order feature interactions between modalities. Jia et al. [44] was inspired by the SimSiam [43] framework and adopted a predictive pretext task to avoid reliance on negative samples.

III. Methodology

In this section, we will elaborate on our ternary modal contrastive learning method, which involves segmenting the hyperspectral modality and conducting contrastive learning between it and the LiDAR modality. The proposed model architecture, as illustrated in Fig. 1, is divided into two main parts. The top half details the ternary modal contrastive learning framework. This process is designed to leverage the unique properties of each modality to enhance the discriminative power of the learned features. The bottom half of Fig. 1 demonstrates the fine-tuning framework that enhances the model's generalization performance through a data augmentation method and multilevel fusion network.

A. Multimodality Contrastive Learning Networks

The efficacy of land cover classification via single-modality frequently does not measure up to the outcomes achieved through multimodal fusion strategies that integrate LiDAR data. This discrepancy in performance primarily arises from the multimodal approaches' capacity to integrate complementary insights sourced from disparate sensors. In this context, contrastive learning has demonstrated significant superiority in processing multimodal data by enhancing the model's discernment of the parallels and distinctions across data derived from various modalities by facilitating a profound integration of modal information, thereby substantially elevating the model's generalization capabilities and robustness. However, HSI blocks may contain pixels belonging to different classes

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:37:32 UTC from IEEE Xplore. Restrictions apply.

than the target pixel, and incorporating features from these regions can adversely affect classification accuracy. Therefore, we have additionally integrated a spectral modality to enhance the feature representation of the target pixel. The detailed steps are as follows.

In this section, we detail our ternary modal contrastive learning approach, which segments the HSI modality and conducts contrastive learning with the LiDAR modality. Assuming we have hyperspectral data $X_H \in \mathbb{R}^{W\times H\times C}$ and LiDAR data $X_L \in \mathbb{R}^{W\times H}$, where W and H represent the width and height of the HSI, respectively, and C represents the number of bands. First, we separate the HSI into its modalities; the spectral modality of a pixel to be classified is $x_m^i \in \mathbb{R}^{1\times 1\times C}$, $i \in \{1, 2, \ldots, N\}$. Due to the properties of HSIs, where different materials might have similar spectral features at certain bands (same spectrum different materials), stacking multiple bands can interfere with spatial information. Therefore, we choose a spatial block of size $s \times s$ centered around x_m^i and select a band from this block to represent the spatial modality $x_a^i \in \mathbb{R}^{s\times s\times 1}$. To enhance the diversity of spatial samples, we adopt a random selection strategy, which gradually covers all spatial information as the number of iterations increases. For the LiDAR data, we similarly select a spatial block of size $s \times s$ at the same location as x_m^i to obtain the elevation modality $x_l^i \in \mathbb{R}^{s\times s\times 1}$.

To enable the model to learn high-level semantic features and reduce noise interference between modalities, we perform random cutout masking on the three modalities. However, experiments showed that masking all three modalities simultaneously made it difficult for the model to fit, so we opted for a primary-auxiliary information masking strategy. When the spectral modality is masked, the spatial and elevation modalities are not cut out, and vice versa.

Subsequently, these three modalities are inputted into spatial, spectral, and LiDAR feature extractors f_a, f_m, and f_l, respectively, to obtain corresponding features f_a^i, f_m^i, and f_l^i. Considering the complexity of each modality and to improve network training efficiency, we use both 1-D and 2-D convolutions. The network structure, as shown in Fig. 1, is composed of three convolutional blocks. It starts with 1×1 convolutions to capture high-dimensional information of the modality, followed by 3×3 convolutional blocks to capture local spatial features and ends with a max-pooling layer to increase the receptive field and filter the features from the convolutional layers. The number of kernels in each convolutional block is 128, 256, and 512, respectively. The features f_a^i, f_m^i, and f_l^i are then projected into a high-dimensional space through a nonlinear transformation network to obtain representations z_a^i, z_m^i, and z_l^i. Our network's learning objective is to learn the similarity between different modalities of the same sample and the dissimilarity between different samples across modalities. Thus, we designed three cross-modal contrastive losses: the space-spectral contrastive loss, the LiDAR-spectrum contrastive loss, and the space-LiDAR contrastive loss. The formulas are as follows:

$$l_{ss} = -\log \frac{\exp\left(\text{sim}\left(z_a^i, z_m^i\right)/\tau\right)}{\sum_{k=1}^{N_b} I_{k\neq i} \exp\left(\text{sim}\left(z_a^i, z_m^k\right)/\tau\right)} \tag{1}$$

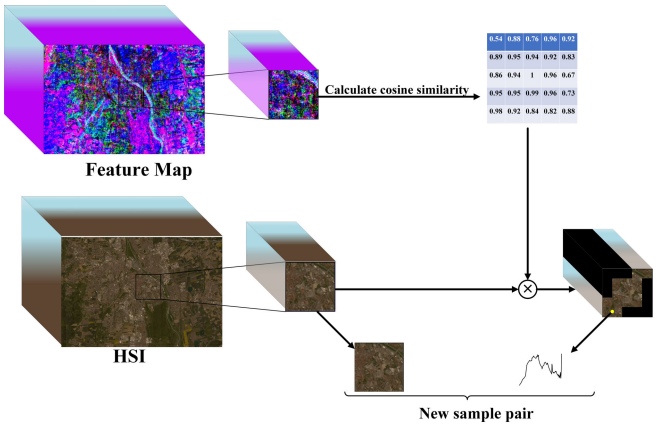


Fig. 2. Spectral feature similarity driven data augmentation.

$$l_{ls} = -\log \frac{\exp\left(\text{sim}\left(z_l^i, z_m^i\right)/\tau\right)}{\sum_{k=1}^{N_b} I_{k\neq i} \exp\left(\text{sim}\left(z_l^i, z_m^k\right)/\tau\right)} \tag{2}$$

$$l_{sl} = -\log \frac{\exp\left(\text{sim}\left(z_a^i, z_l^i\right)/\tau\right)}{\sum_{k=1}^{N_b} I_{k\neq i} \exp\left(\text{sim}\left(z_a^i, z_l^k\right)/\tau\right)}. \tag{3}$$

In the above equations, N_b is the batch size for each iteration, and τ adjusts the smoothness of the logistic distribution. I is an indicator function that equals 1 when the condition is met and 0 otherwise. For various datasets with distinct data characteristics, we employ different loss functions to update the model. For datasets such as Houston2013, Trento, and MUUFL, where there are significant chemical property differences between categories, we solely utilize the l_{ss} and l_{ls} to update the parameter. Conversely, for the Augsburg dataset, which contains categories with chemically similar substances, we employ these three loss functions. A detailed analysis of this design choice can be found in Section IV-E. The parameters of the three feature extractors are saved for use as initial parameters in downstream classification tasks.

B. Data Augmentation for HSI

Data augmentation, through the introduction of perturbations, serves to augment the diversity of the sample set, thereby enhancing the model's capacity for generalization. Traditional methods of random augmentation, which predominantly alter spatial information, prove suboptimal for the task of point classification within the realm of HSI and LiDAR. As illustrated in Fig. 2, drawing upon the inherent neighborhood similarity characteristic of HSI, we propose an innovative data augmentation technique designed to expand the variance coverage and enhance the predictive capacity for unseen samples under the condition of using a limited number of labeled samples. The procedural details are articulated as follows.

First, features f_a, f_m, and f_l obtained from an unsupervised pretrained model are summed to obtain the feature map $f_{\text{img}} \in \mathbb{R}^{H\times W\times C_2}$, where C_2 is the feature channel dimension. Let $\{x_a^t, x_m^t, x_l^t, y^t\}, t \in \{1, 2, 3, \ldots, M\}$ be the training set, with M being the number of training set samples. x_a^t, x_m^t, and x_l^t represent the spatial, spectral, and elevation modalities to be classified, respectively. y^t is the manually annotated label. A feature block $f_e \in \mathbb{R}^{p\times p\times C_2}$ of size $p \times p$, centered around

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:37:32 UTC from IEEE Xplore. Restrictions apply.

x_m^t, is extracted on f_{img}. This procedure is replicated on the HSI $X_H \in \mathbb{R}^{W\times H\times C}$, yielding an image block $f_p \in \mathbb{R}^{p\times p\times C}$. To mitigate the classification interference caused by features from pixels of disparate classes relative to the target pixel, we compute the cosine similarity between each feature vector within the block f_e and the central feature vector. Similarities exceeding a predefined threshold p are assigned a value of 1, while those equal to or below p are set to 0, thereby generating a similarity matrix $M_s \in \mathbb{R}^{s\times s}$. $f_p[:,:,k]$ represents the $p \times p$ image of the kth band in the image block f_p. The refined image block f_n, deleted of hetero-class pixels, is delineated as

$$f_n[:,:,k] = M_s \circ f_p[:,:,k]. \tag{4}$$

Here, $\circ$ signifies the Hadamard product. From the refined image block f_n, a spectral band $x_m^{t'}$ is randomly selected to represent the new spectral modality, with operational specifics articulated in the following equation:

$$x_m^{t'} = \begin{cases} x_m^{t'}, & \text{if } \|x^{t'}\| > 0 \\ x_m^t, & \text{otherwise.} \end{cases} \tag{5}$$

The spatial and elevation modalities are maintained without alteration.

C. Multilevel Fusion Networks for Muti-Modality Model

As the network level goes deeper, it evolves from capturing elementary detailed features to extracting higher-level semantic information. The fusion of these layers of information is crucial for enhancing the model's ability to globally understand the data [45]. Particularly in the realm of synergistic classification of hyperspectral and LiDAR data, HSIs may exhibit noise or incompleteness in detail features when affected by weather conditions. In such scenarios, LiDAR data can efficaciously augment the noise-afflicted hyperspectral imagery by leveraging its robust low-level detail features, thereby maintaining the model's inferential performance. This cross-modal, multilevel feature fusion strategy not only strengthens the model's overall understanding of diverse data types but also improves classification accuracy and robustness in complex environments.

Drawing on the concept from [46], we designed a multimodal, multilayer feature fusion network that allows the model to integrate features across different scales, as shown in the bottom part of Fig. 1. The complexity of the feature fusion module necessitates a large sample size for fitting, which is constrained by the amount of labeled data available. To address this, we use simple linear layers to project the shallow features from Block1 and the mid-level features from Block2 to the same dimensionality as features f_a, f_m, and f_l. Considering the potential curse of dimensionality associated with high-dimensional spaces, for each modality, we directly sum the shallow, mid-level, and high-level features to obtain features $f_{\text{in}A}$, $f_{\text{in}M}$, and $f_{\text{in}L}$, and then concatenate these features to form the final feature f, which is fed into a classifier for categorization. The model parameters are updated using a cross-entropy function, as illustrated below

$$L_c = -\sum_{t=1}^{M} y^t \log\left(\frac{\exp\left(W_i^T \times f\right)}{\sum_{j=1}^{K}\exp\left(W_j^T \times f\right)}\right) \tag{6}$$

where K is the number of categories to be classified in the dataset.

IV. Experiments

To comprehensively evaluate our proposed method, we conducted a series of experiments utilizing four multisensor datasets: Houston2013, MUUFL Gulfport, Trento, and Augsburg. Initially, we delineate comprehensive details regarding these datasets, subsequently offering an elucidation of our experimental setting. Extensive ablation experiments were performed to validate the effectiveness of each module. Finally, our classification results were compared with SOTA methods from recent years to demonstrate the superior classification performance of our approach.

A. Dataset Description

1) Houston2013: The Houston2013 dataset encompasses imagery obtained via the ITRES CASI-1500 sensor, targeting the University of Houston campus and its adjacent urban environs. This sensor, operational across the spectral wavelength range of 0.38–1.05 μm, acquires data across 144 spectral bands. The dataset presents both hyperspectral and LiDAR images, each with dimensions of 349 by 1905 pixels. It comprises a total of 15 029 pixels to classify, segmented into 15 unique classes for the purpose of classification.

2) MUUFL: The MUUFL Gulfport dataset was acquired utilizing the Reflective Optics System Imaging Spectrometer (ROSIS) sensor, stationed at the Gulf Park campus of the University of Southern Mississippi. This sensor delineates the picture in the wavelength interval from 0.375 to 1.05 μm, spanning 72 spectral bands. Owing to the presence of noise interference, eight of these bands were deemed unsuitable for analysis and were subsequently omitted, resulting in a final tally of 64 bands employed in the research endeavors. The dataset features both hyperspectral and LiDAR imagery, each with a resolution of 320 $\times$ 220 pixels. Encompassing a total of 55 337 pixels, the dataset is stratified into 11 distinct classes for classification purposes, including, but not limited to, trees, roads, and water.

3) Trento: The Trento dataset's hyperspectral imagery was procured via the AISA Eagle sensor, focusing on a rural locale situated to the south of Trento city. Concurrently, the LiDAR imagery was garnered utilizing the Optech ALTM3100EA sensor, characterized by a grid size of 1. The captured HSIs are comprised of 63 spectral bands, with the spatial dimensions delineated as 600 $\times$ 166 pixels. Within this dataset, 30 214 pixels have been earmarked for classification endeavors, these pixels being systematically segregated into six distinct categories.

4) Augsburg: The Augsburg dataset was captured in the city of Augsburg. LiDAR data was obtained through the dlr-36 system, while the HSIs were acquired using the DAS-EOC HySpex sensor. This sensor captures data in the wavelength

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:37:32 UTC from IEEE Xplore. Restrictions apply.

TABLE I
TRAINING AND TEST SAMPLE NUMBERS FOR HOUSTON2013, TRENTO, MUUFL, AND AUGSBURG DATASETS

No.	Houston2013 dataset			MUUFL dataset			Trento dataset			Augsburg dataset		
	Categories	Training	Test	Categories	Training	Test	Categories	Training	Test	Categories	Training	Test
C1	Healthy Grass	10	1241	Trees	10	23236	Apple Trees	10	4024	Forest	10	13497
C2	Stressed Grass	10	1244	Mostly Grass	10	4260	Buildings	10	2893	Residential Area	10	30319
C3	Synthetic Grass	10	687	Mixed Ground Surface	10	6872	Ground	10	469	Industrial Area	10	3841
C4	Trees	10	1234	Dirt and Sand	10	1816	Woods	10	9113	Low Plants	10	26847
C5	Soil	10	1232	Road	10	6677	Vineyard	10	10491	Allotment	10	565
C6	Water	10	315	Water	10	456	Roads	10	3164	Commercial Area	10	1635
C7	Residential	10	1258	Buildings Shadow	10	2223				Water	10	1520
C8	Commercial	10	1234	Buildings	10	6230						
C9	Road	10	1242	Sidewalk	10	1375						
C10	Highway	10	1217	Yello Curb	10	173						
C11	Railway	10	1225	Cloth Panels	10	259						
C12	Parking Lot1	10	1223									
C13	Parking Lot2	10	459									
C14	Tennis Court	10	418									
C15	Running Track	10	650									
Total		150	14879		110	53577		60	30154		70	78224

TABLE II
ABLATION STUDIES ON DIFFERENT DATASETS. ALL THE BEST ACCURACIES ARE IN BOLD

Pretrain	Encoder	Spectral selection	Multi-Level fusion	Houston2013	Muufl	Trento	Augsburg
-	✓	-	-	80.89	75.28	97.04	60.85
✓	✓	-	-	92.53	81.62	97.62	87.83
✓	✓	✓	-	93.13	81.93	97.94	88.45
✓	✓	✓	✓	**93.80**	**83.04**	**98.58**	**88.48**

range of 0.4–2.5 μm, encompassing 180 spectral bands. The dimensions of the HSIs are 332 × 485 pixels. A total of 78 294 pixels are utilized for classification in this dataset, covering seven different land cover categories.

The number of training samples and testing samples for each land-cover category in the four datasets mentioned above is shown in Table I.

B. Experimental Setup

For this experiment, we chose the Pytorch framework, running on the Ubuntu 20.04 LTS operating system with an Nvidia GeForce RTX 3090 GPU. During the pretraining phase of contrastive learning, we selected 80% of the unlabeled data for model training. The patch size was set to 9, and the cutout size was set to 40% of the dataset. The batch size was 512, with a total of 20 iterations, and the learning rate was set to 3e-4.

In the fine-tuning stage for the classification task, the chosen feature block size was 3, with a threshold set at 0.9. Ten samples per class were selected. The feature extractor's learning rate was set to 3e-5, and the classifier's learning rate was 3e-4. The batch size was 16, with 200 epochs of iteration. We evaluated the algorithm's performance using three metrics: overall accuracy (OA), average accuracy (AA), and the Kappa coefficient.

C. Ablation Study

To validate the effectiveness of our proposed model, we undertook exhaustive ablation studies targeting each constituent component. Table II demonstrates the classification accuracy obtained by applying different modules. The first column indicates the use of multimodal contrastive learning for pretraining, the second column indicates the use of neighborhood spectral vector selection methods, and the penultimate column represents the use of multilevel feature fusion network methods, with a ✓denoting the application of that module. Noteworthy enhancements in classification accuracy were discernible across the Houston2013, MUUFL, and Augsburg datasets, attributable to the incorporation of contrastive learning, yielding incrementations of 11.64%, 6.34%, and 26.98%, respectively. The Trento dataset, due to its notable intraclass consistency, achieved high classification accuracy even without pretraining but still saw a 0.58% accuracy improvement through contrastive learning pretraining. This demonstrates that contrastive learning, by bringing positive sample pairs closer and pushing negative pairs further apart, can effectively extract meaningful feature representations from data. These features are highly effective for downstream classification tasks. Utilizing the neighborhood spectral vector data augmentation algorithm, as shown in Table II, significantly improved the classification performance of each dataset, proving that this data augmentation method can effectively enhance data diversity. Employing multilevel networks facilitates the provision of a more comprehensive feature set, wherein the initial layers capture fundamental geometrical shapes inherent within the data, and the subsequent, deeper layers discern intricate patterns. Owing to the intricate land cover composition of the MUUFL dataset, an enhancement in the precision of 1.11% was realized through the integration of multilevel feature fusion.

In a bid to further verify the robustness of our methodological proposition, Fig. 3 presents a t-distributed stochastic neighbor embedding (t-SNE) visualization of disparate features.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:37:32 UTC from IEEE Xplore. Restrictions apply.

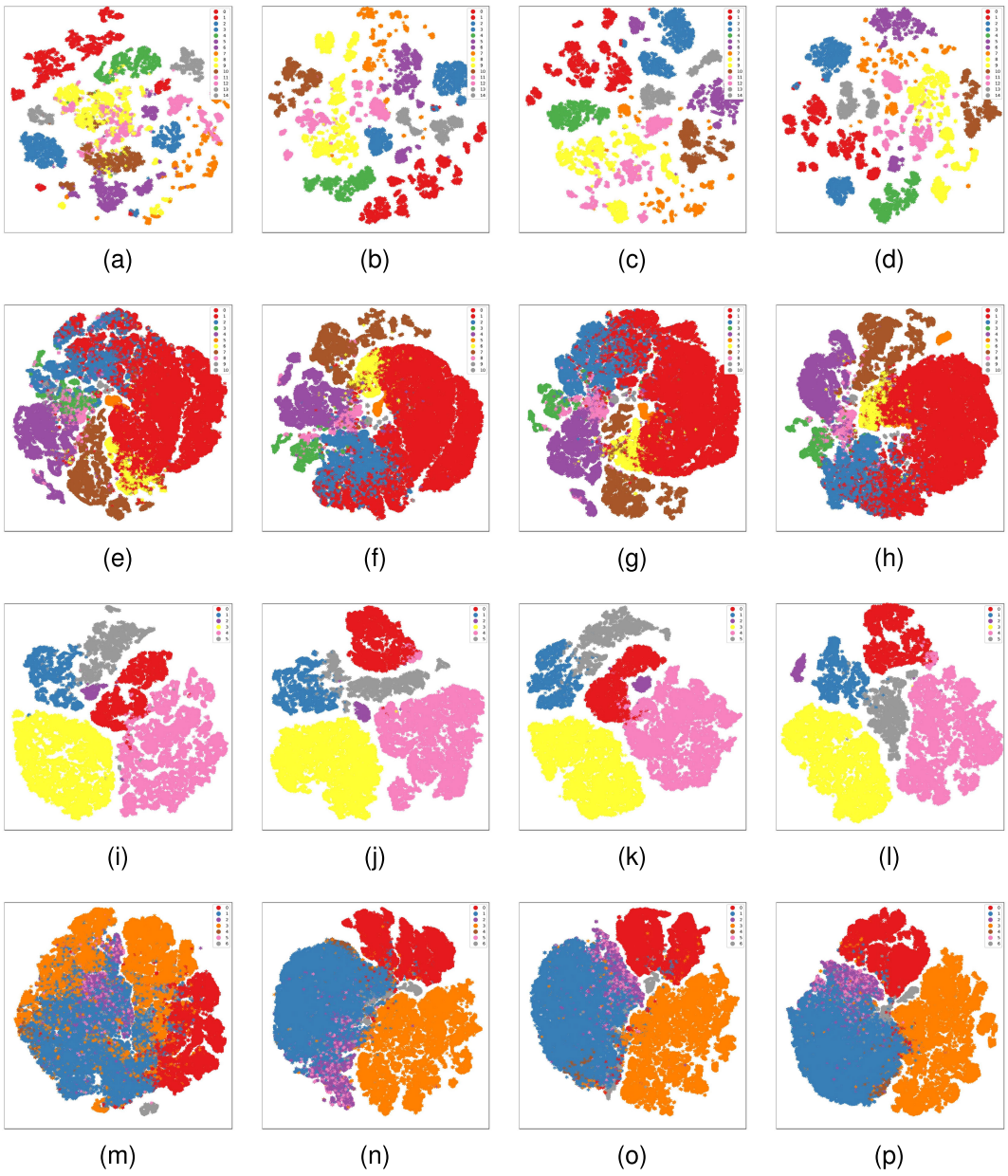


Fig. 3. Feature visualization of encoders for HSI and LiDAR classification on four datasets. (a)–(d) Results on the Houston2013 dataset. (e)–(h) Results on the MUUFL dataset. (i)–(l) Results on the Trento dataset. (m)–(p) Results on the Ausburg dataset. The first column represents training the model directly using labeled data. The second column indicates pretraining with contrastive learning. The third column involves the use of a spectral vector selection data augmentation algorithm, and the last column employs a multifusion network.

Postapplication of contrastive learning, an expansion in the intercategory sample distances is observable, whilst intracategory samples exhibit enhanced compactness. Concurrently, the multilevel network's application precipitates a significant diminution in the prevalence of outlier points.

D. Analysis of the Patch Size of Neighborhood Spectral Vectors

The selection of feature block size impacts the reliability of samples. An increase in the dimensions of the feature block precipitates a diminution in the semantic congruence between the chosen spectral and spatial modalities, which can negatively affect classification. As illustrated in Fig. 4, all four datasets show that when the feature block size is set to 3, the improvement in classification results is the best. Owing to its spatial uniformity, the Trento dataset exhibits a progressive decrement in accuracy with the augmentation of block size. Conversely, the MUUFL dataset, characterized by its intricate land cover dispersion, experiences a more pronounced degradation in classification performance as block size enlarges.

E. Contrastive Learning With Three Views

In this section, we attempt to discuss how to design the model's loss function when dealing with three views, aiming to strike an optimal balance between the model's efficacy and computational efficiency. The core of contrastive learning is to maximize the mutual information between variables, such as

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:37:32 UTC from IEEE Xplore. Restrictions apply.

TABLE III
IMPACT OF DIFFERENT LOSS FUNCTION STRATEGIES ON THE ACCURACY OF THE HOUSTON2013, MUUFL, TRENTO, AND AUGSBURG DATASETS

	Houston2013		MUUFL		Trento		Augsburg	
	Core loss	Full loss	Core loss	Full loss	Core loss	Full loss	Core loss	Full loss
OA	**93.80**	92.19	**83.04**	82.59	**98.58**	97.42	85.56	**88.49**

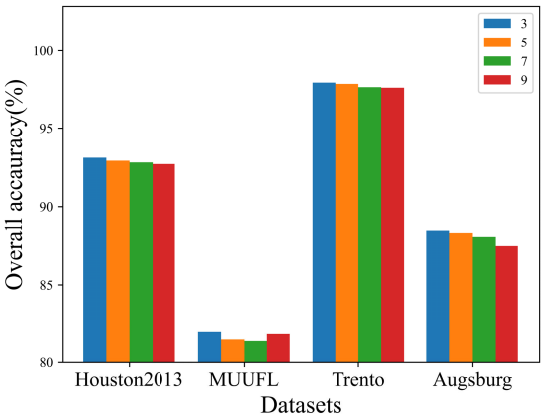


Fig. 4. Impact of different feature block size selections on classification accuracy.

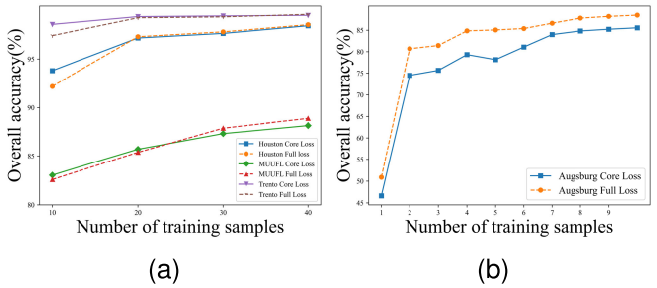


Fig. 5. Accuracy of full contrastive learning and core contrastive learning across four datasets at various sample sizes. (a) Houston, MUUFL and Trento dataset. (b) Augsburg dataset.

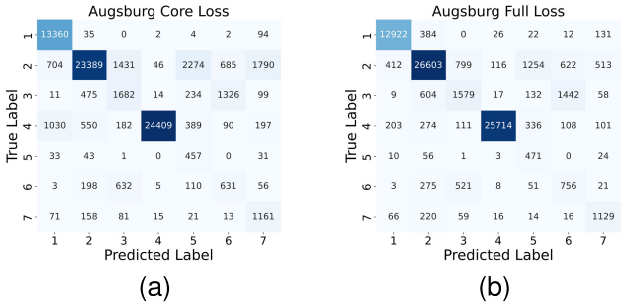


Fig. 6. Confusion matrices for the Augsburg dataset under full contrastive learning and core contrastive learning.

obtaining two different perspectives of the same image through different color disturbances, to learn representations that are invariant to these view changes [38]. In reality, maximizing mutual information is undoubtedly very challenging, and researchers usually maximize its lower bound by minimizing the InfoNce loss, indirectly maximizing the lower bound of mutual information. The InfoNce loss function is defined as follows:

$$L_{\text{infoNCE}} = -\log \frac{\exp(\text{sim}(a,b)/\tau)}{\sum_{b\in B}\exp(\text{sim}(a,b)/\tau)} \tag{7}$$

where B is a set consisting of a positive sample b and $B-1$ negative samples from the distribution of samples we have. Our losses are, respectively, the space-spectral contrastive loss. The implementation of the formula is as shown

$$l_{ss} = -\log \frac{\exp\left(\text{sim}\left(z_a^i, z_m^i\right)/\tau\right)}{\sum_{k=1}^{N_b} I_{k\neq i}\exp\left(\text{sim}\left(z_a^i, z_m^k\right)/\tau\right)}. \tag{8}$$

Minimizing the space-spectral contrastive loss can be similarly considered as minimizing a symmetric InfoNCE loss. Therefore, we view the spectral and spatial modalities as different perspectives of space-spectral pairs, enabling the model to learn mutual information between the spatial and spectral modalities. Similarly, the LiDAR-spectrum contrastive loss is considered as learning the mutual information between the LiDAR and spectral modalities.

Regarding the architecture of contrastive learning models for three modalities, we propose two strategies: 1) full contrastive learning: entails executing contrastive learning across every pair of modalities. 2) Core contrastive learning: select the modality containing the most information as the core modality to perform contrastive learning with other modalities. Although full contrastive learning facilitates the exploration of complementary information amongst the model's different modalities, but its computational complexity is higher, and it is prone to overfitting with a scant number of samples for fine-tuning. Our comparative analysis of these contrastive learning paradigms across four datasets, as shown in Table III revealed that for the Houston2013, MUUFL, and Trento datasets, the precision of core contrastive learning supersedes that of full contrastive learning with ten samples per class. As depicted in Fig. 5, as the number of samples increases, the accuracy of full contrastive learning gradually surpasses that of core contrastive learning. However, on the Augsburg dataset, even when the sample size is 1, the accuracy of full contrastive learning is higher than that of the core contrastive learning method. We visualized the classification confusion matrix, as shown in Fig. 6, and found that misclassifications mostly occur between the first and fourth classes, and between the second and fifth classes. Based on the category information and ground truth, it can be known that the two have similar chemical substances, and merely mining the mutual information of the spectral and spatial modalities, and the spectral and LiDAR modalities might be insufficient. As we mentioned before, the spatial information provided by hyperspectral comes from the light reflection of specific bands, while the spatial information provided by LiDAR comes from the height difference between different land covers and the sensor. The exploration of complementary information between the two plays an important role in the classification of land covers with similar chemical substances leading to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:37:32 UTC from IEEE Xplore. Restrictions apply.

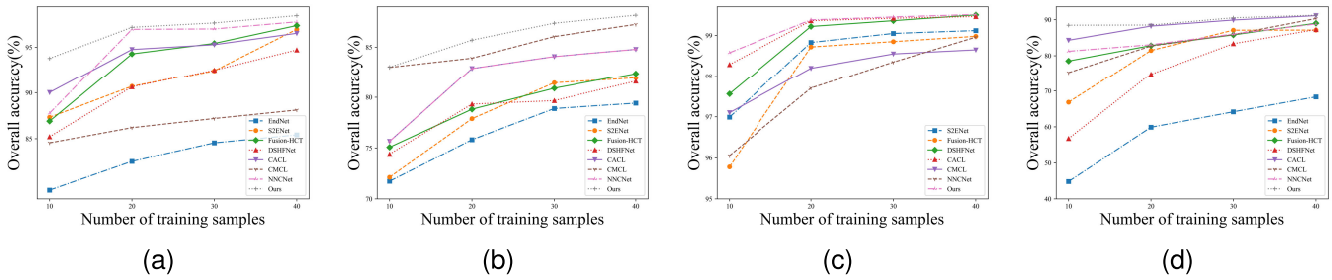


Fig. 7. Classification results in terms of OA values under different training samples on four datasets. (a) Houston2013. (b) MUFFL. (c) Trento. (d) Augsburg.

similar spectra. Therefore, we deduce that in scenarios with a limited corpus of manually annotated samples for fine-tuning, using core contrastive learning can usually achieve better accuracy. However, when there are chemical substances extremely similar and land cover distributions are close, using full contrastive learning can ensure better accuracy. Moreover, as the number of samples available for fine-tuning increases, full contrastive learning achieves higher accuracy than core contrastive learning.

F. Robustness and Model Generalization

We conducted an empirical evaluation of the OA performance of our proposed methodology compared to existing methods across four datasets with varying sample volumes, as depicted in Fig. 7. Our model demonstrated minor fluctuations in accuracy across different training sample sizes and consistently achieved the highest precision on all datasets regardless of sample volume. This suggests that our approach exhibits low sensitivity to the number of training samples, illustrating superior robustness and generalization capabilities. On the contrary, the supervised Fusion-HCT method exhibited significant accuracy variations as the sample size increased. This variability arises because supervised methods depend on ample labeled samples to learn the general patterns of data distribution, and when samples are scarce, the model is prone to overfitting, capturing specific features of the limited data pool. Conversely, our methodology leverages extensive pretraining on unlabeled samples to mine features from the sample space distribution, ensuring robust feature extraction. Similarly, like our approach, CMCL and NNCNet employ pretraining on unlabeled samples, which contributes to the minimal variability in model accuracy with changing sample volumes.

G. Comparison of Classification Results

In this section, we compare our method with baseline methods EndNet, supervised methods S2Enet, Fusion-HCT, DSHFNet, semi-supervised methods CMCL, contrastive learning methods CMCL, and NNCNet, running them with their respective papers' default optimal parameters to obtain results.

1) EndNet: Utilizes a deep encoder–decoder network architecture for the classification of joint hyperspectral and LiDAR data.
2) S2ENet: Employs a cross-modal information interaction spatial–spectral enhancement module to facilitate interaction and understanding among multimodal data.
3) Fusion-HCT: Composed of hierarchical convolutional networks and transformers to extract hyperspectral spectral–spatial information and LiDAR elevation information.
4) DSHFNet: Proposes a dynamic scale hierarchical fusion network to integrate small-scale features into larger scales by threshold judgment, addressing the issue of unreliability in single-scale features.
5) CACL: Trains an adversarial network with dual generators and discriminators based on coupled adversarial learning.
6) CMCL: Treats HSI and LiDAR as different perspectives of the same object for contrastive learning and adopts a dual fine-tuning strategy to transfer the extracted features.
7) NNCNet: Designs a bilinear attention module to interact with high-order features between HSI and LiDAR.

Tables IV–VII display the results of our proposed method compared to these methods on the Houston2013, Muufl, Trento, and Augsburg datasets for OA, AA, and Kappa. In this table, the best results are highlighted in bold. Especially on the Houston2013 and Augsburg datasets, our method is, respectively, 3.77% and 4.18% higher than the SOTA methods.

The Houston dataset, due to its considerable sample volume and sparse distribution of labeled samples, does not facilitate the deployment of DSHFNet's multiscale capabilities effectively. Conversely, S2Enet, which integrates a spatial–spectral enhancement module, demonstrates superior competitiveness, achieving a classification accuracy that is 2.12% higher than that of DSHFNet. Our method outperforms other competitive methods like CACL in evaluation metrics by 3.77%, 3.48%, and 4.09%, respectively. Our method performs significantly better in the water and road categories. Fig. 8 illustrates the visualization of various methods on the Houston2013 dataset, showing our method still performs excellently in areas covered by clouds and fog on the right side of the dataset, demonstrating our method's ability to fully exploit LiDAR information and maintain excellent classification performance in areas where HSI information is compromised.

The MUUFL dataset, with its densely labeled samples and regions of irregularly shaped categories, posits that the utilization of spatial information may introduce unnecessary interference categories. Thus, EndNet, which merely reconstructs spectral pixels, and S2Enet, which uses spatial blocks in a supervised method, are only 0.39% apart in performance. Fig. 9 visualizes the prediction results of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:37:32 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
CLASSIFICATION PERFORMANCE OF DIFFERENT MODELS ON THE HOUSTON2013 DATASET

Class No.	EndNet	S2ENet	Fusion-HCT	DSHFNet	CMCL	CACL	NNCNet	Ours
1	79.13	85.66	**88.99**	85.65	83.08	85.41	78.00	87.38
2	92.77	96.95	90.19	95.90	96.00	93.65	96.86	**97.80**
3	99.71	99.56	98.95	98.10	95.01	93.45	99.71	**100**
4	96.03	92.71	95.93	94.49	**98.84**	89.63	87.12	98.82
5	79.38	99.35	99.89	**100**	100	100	99.11	98.31
6	93.33	81.59	94.41	93.65	97.20	88.57	97.14	**99.63**
7	80.37	81.00	83.39	66.69	**93.28**	90.30	78.22	91.70
8	80.39	**91.33**	71.38	75.36	88.13	74.96	73.58	86.78
9	66.26	72.71	67.42	36.63	80.49	79.31	59.50	**85.94**
10	49.22	47.25	83.40	89.73	57.55	82.74	**93.18**	92.26
11	86.04	94.37	82.14	94.37	79.05	98.04	98.69	**98.99**
12	74.73	90.76	84.20	88.06	52.60	**92.40**	91.58	86.41
13	43.79	99.78	93.03	97.82	92.49	**100**	96.51	99.01
14	72.73	100	100	97.61	84.78	**100**	99.76	97.76
15	99.08	**100**	100	100	98.73	100	99.38	99.89
OA	79.44	87.33	86.91	85.21	84.55	90.03	87.77	**93.80**
AA	79.53	88.87	88.89	87.61	86.48	91.23	89.89	**94.71**
Kappa	86.31	84.02	86.64	84.02	83.24	89.22	86.78	**93.31**

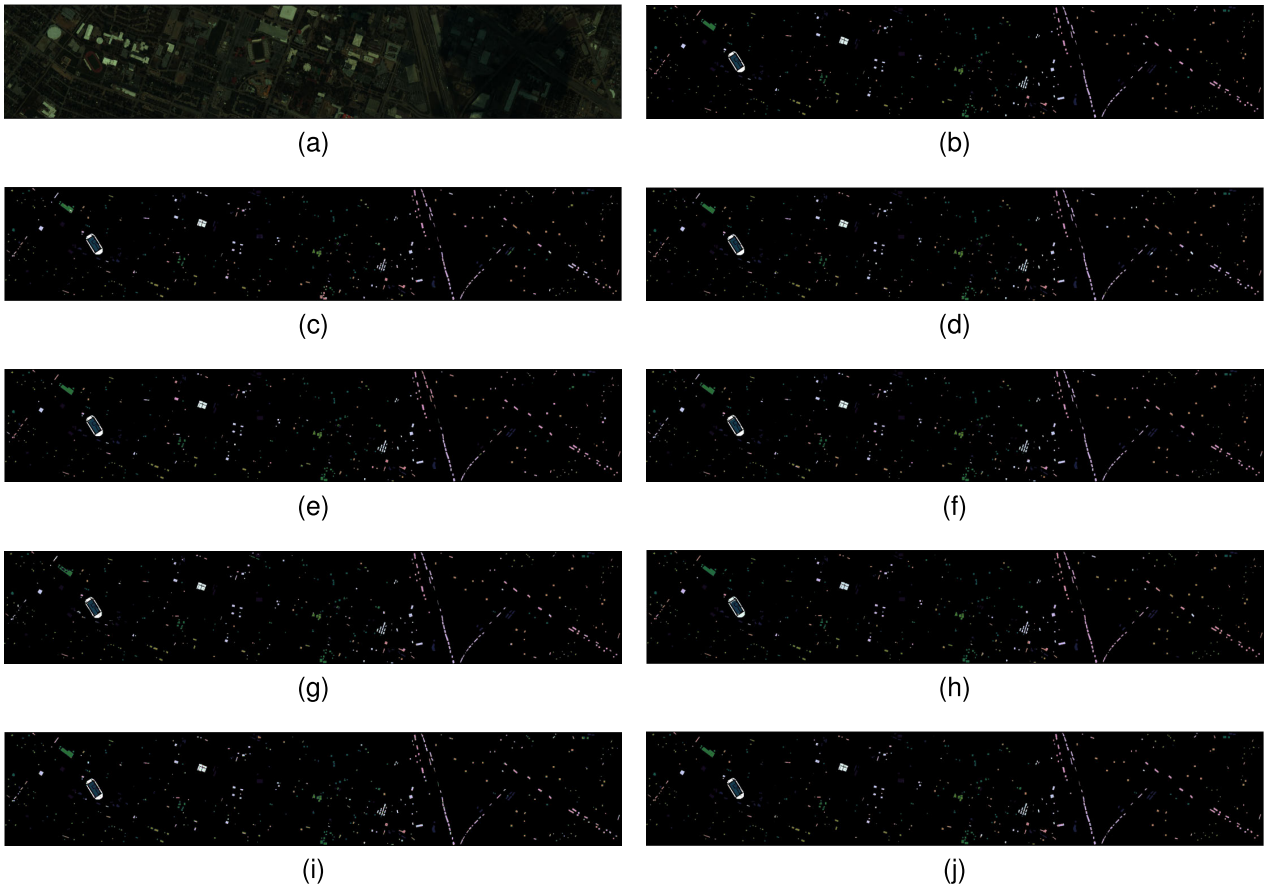


Fig. 8. Classification maps of each algorithm on the Pavia University dataset. (a) Pseudo-color. (b) Ground-truth. (c) EndNet (79.44%). (d) S2ENet (87.33%). (e) Fusion_HCT (86.91%). (f) DSHFNet (85.21%). (g) CMCL (84.55%). (h) CACL (90.03%). (i) NNCNet (87.77%). (j) Proposed method (93.80%).

various methods on MUUFL, showing S2Enet's high classification accuracy for regular regions through the spatial–spectral enhancement module, but a high probability of misclassification for irregular areas. Our method, by mining the mutual information between spatial, elevation, and spectral domains, reduces the influence of nontarget pixels in spatial blocks on classification, performing better than other methods.

The Trento dataset's different categories are usually in separate areas with high spatial homogeneity and low intraclass heterogeneity. Furthermore, the reduced number of categories lessens the requisite for diversity among negative

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:37:32 UTC from IEEE Xplore. Restrictions apply.

TABLE V
CLASSIFICATION PERFORMANCE OF DIFFERENT MODELS ON THE MUUFL DATASET

Class No.	EndNet	S2ENet	Fusion-HCT	DSHFNet	CMCL	CACL	NNCNet	Ours
1	69.09	73.58	82.70	74.91	**85.03**	86.35	80.31	83.83
2	81.81	79.06	72.81	**85.73**	84.08	54.48	85.05	84.11
3	40.64	40.79	50.89	36.79	56.60	51.32	50.80	**67.02**
4	55.78	64.98	83.33	87.06	84.22	**90.31**	85.57	88.71
5	86.72	78.57	65.22	84.09	87.58	64.46	65.60	**87.81**
6	98.90	78.95	99.38	98.25	99.87	**100**	98.03	99.04
7	89.20	87.45	81.57	90.55	**97.84**	84.89	76.88	93.58
8	90.88	87.54	84.63	90.42	**97.10**	89.50	81.17	87.92
9	57.75	58.69	46.07	53.09	**74.28**	30.11	41.82	66.47
10	95.38	80.35	76.07	0	82.42	54.34	62.43	**99.31**
11	94.59	95.37	90.27	95.75	**98.99**	95.75	97.30	98.92
OA	71.74	72.13	75.07	74.39	83.01	75.65	74.39	**83.04**
AA	78.25	75.03	75.72	72.42	85.91	72.86	74.99	**86.97**
Kappa	65.18	65.48	68.39	67.96	78.18	68.85	67.60	**78.39**

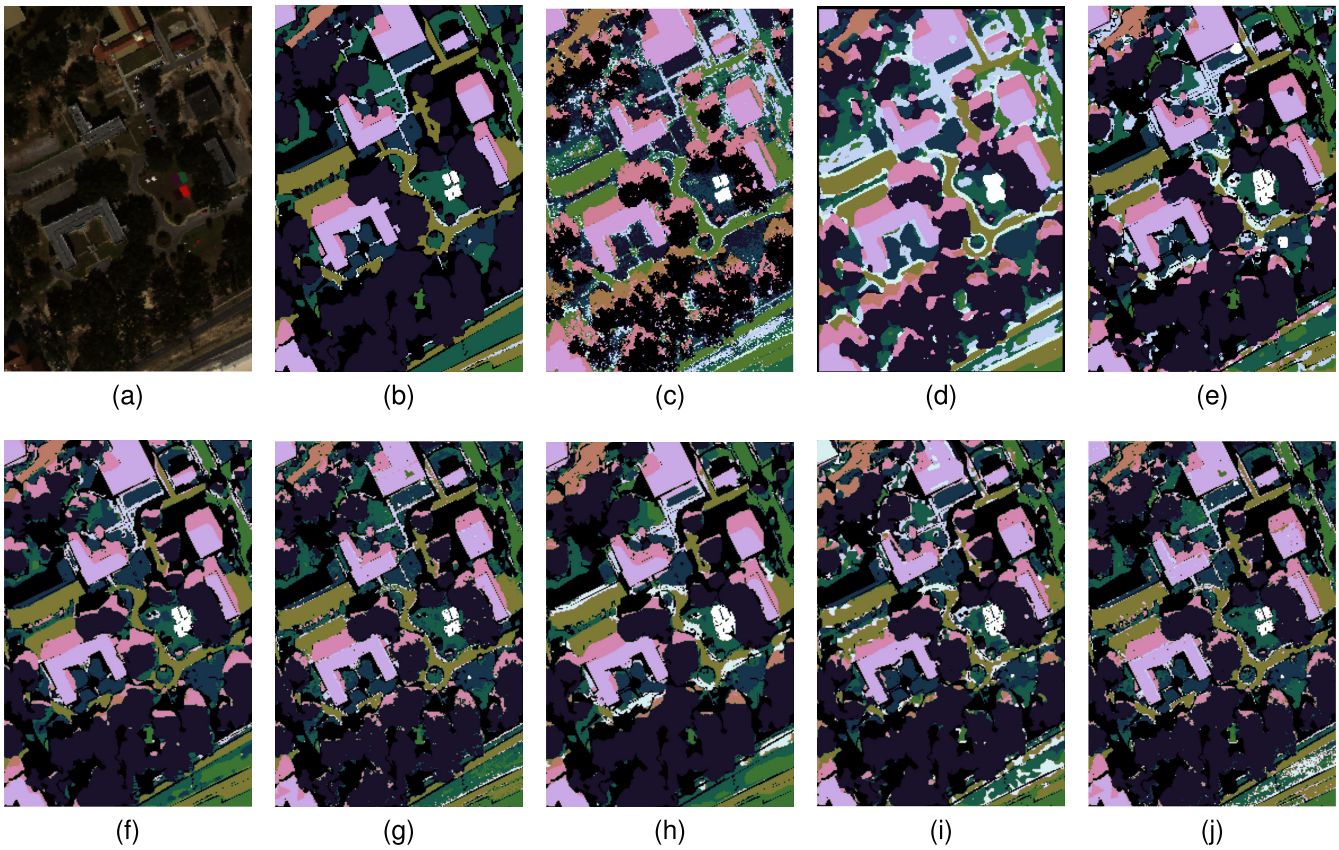


Fig. 9. Classification maps of each algorithm on the Pavia University dataset. (a) Pseudo-color. (b) Ground-truth. (c) EndNet (71.74%). (d) S2ENet (72.13%). (e) Fusion_HCT (75.07%). (f) DSHFNet (74.39%). (g) CMCL (83.01%). (h) CACL (75.65%). (i) NNCNet (74.39%). (j) Proposed method (83.04%).

samples. NNCnet, employing a momentum encoder, falls short of CMCL's dynamic negative sample library updating strategy aimed at preserving feature consistency, resulting in a classification accuracy deficit of 1.05%. Our methodology, predicated on generating novel samples leveraging the neighborhood similarity intrinsic to hyperspectral imagery and enhancing sample diversity, excels beyond the CMCL approach by 1.48%. Fig. 10 illustrating the predictive outcomes of diverse strategies on Trento indicates CACL's frequent misclassifications within the road category, whereas our model, through the formulation of a ternary modal system, secures superior classification outcomes.

The Augsburg dataset has an intricate spatial structure, high spectral resolution, and substantial data redundancy. Most methods perform poorly under a few labeled samples. Fusion-HCT, capitalizing on transformers' long-range information capturing capability, adeptly mines spectral data, securing an accuracy enhancement of 11.66% over the S2Enet method, which relies solely on CNNs. CACL, utilizing a large number of unlabeled samples for adversarial training, performs second only to ours, with our method's

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:37:32 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
CLASSIFICATION PERFORMANCE OF DIFFERENT MODELS ON THE TRENTO DATASET

Class No.	EndNet	S2ENet	Fusion-HCT	DSHFNet	CMCL	CACL	NNCNet	Ours
1	64.56	98.41	96.15	99.83	94.83	99.08	98.21	**99.40**
2	81.61	85.69	96.00	97.92	99.18	**99.76**	92.78	97.82
3	78.89	100	95.05	98.91	87.86	**100**	98.29	97.27
4	93.27	99.45	100	97.89	99.44	**100**	100	99.93
5	69.92	99.56	93.61	99.80	**99.99**	99.04	92.05	98.13
6	84.61	89.48	88.02	85.76	82.61	88.24	**97.85**	95.00
OA	79.06	96.99	95.78	97.56	97.10	98.29	96.05	**98.58**
AA	78.81	95.43	95.16	96.68	93.99	97.69	96.53	**97.93**
Kappa	72.58	95.99	94.43	96.74	96.11	97.72	94.77	**98.07**

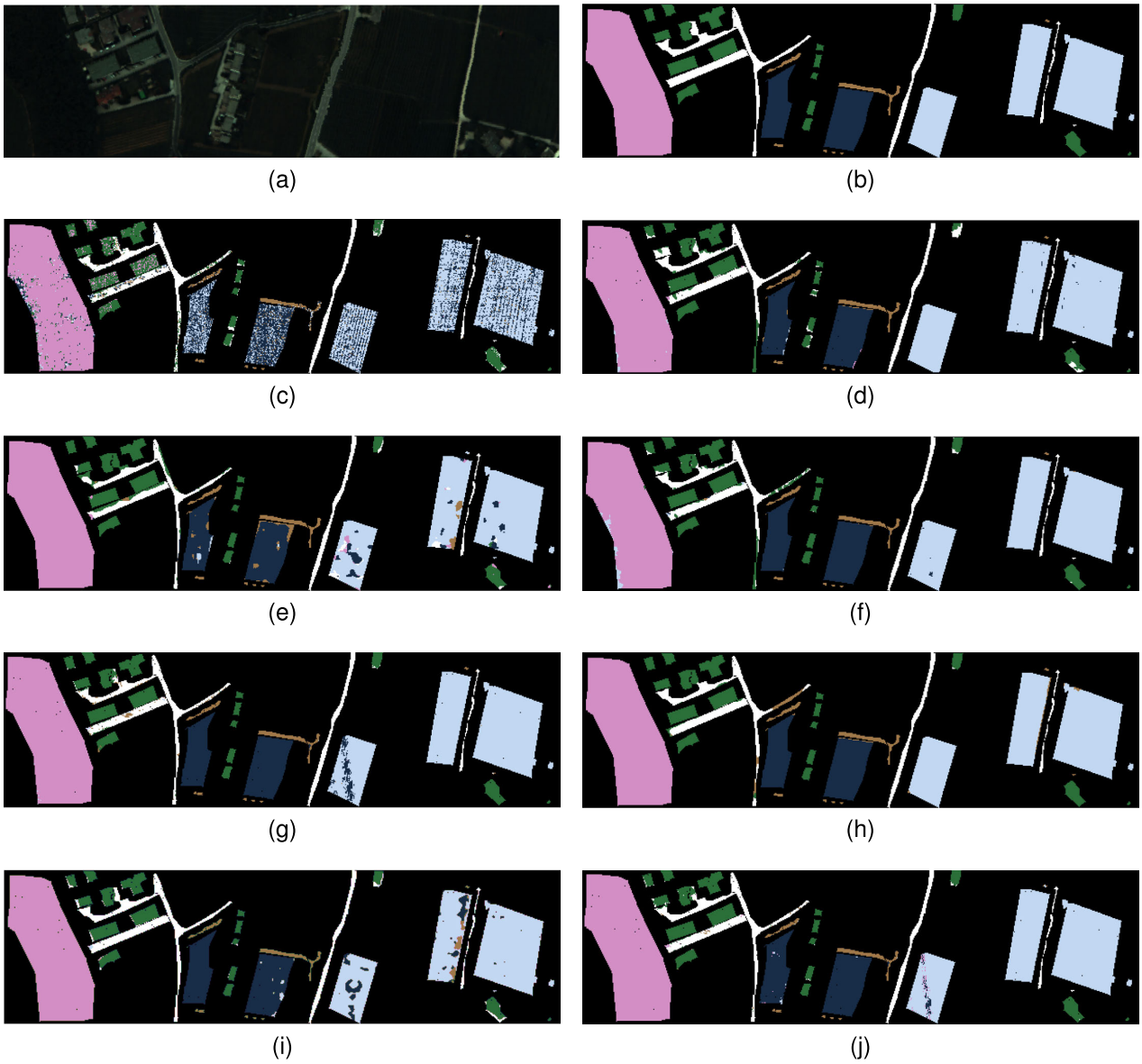


Fig. 10. Classification maps of each algorithm on the Pavia University dataset. (a) Pseudo-color. (b) Ground-truth. (c) EndNet (79.06%). (d) S2ENet (96.99%). (e) Fusion_HCT (95.78%). (f) DSHFNet (97.56%). (g) CMCL (97.10%). (h) CACL (98.29%). (i) NNCNet (96.05%). (j) Proposed method (98.58%).

classification accuracy 4.18% higher than CACL's. Fig. 11 is the classification map of Augsburg, showing our method has fewer regional misclassifications and some point errors, attributable to our segmentation of HSI into spatial and spectral domains and the full exploitation of its spatial modality and LiDAR modality, substantially mitigating the influence of adjacent interference points on classification accuracy.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:37:32 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
CLASSIFICATION PERFORMANCE OF DIFFERENT MODELS ON THE AUGSBURG DATASET

Class No.	EndNet	S2ENet	Fusion-HCT	DSHFNet	CMCL	CACL	NNCNet	Ours
1	79.62	93.79	96.31	**99.30**	94.98	96.48	97.53	97.98
2	35.34	65.35	80.41	42.17	69.20	88.67	76.40	**87.94**
3	34.91	**68.24**	47.99	47.33	34.83	59.75	25.54	38.26
4	37.29	57.43	74.40	51.96	80.82	82.68	88.97	**96.50**
5	92.39	83.36	92.25	**96.81**	88.85	57.35	79.81	55.33
6	37.74	20.37	46.79	51.31	45.02	24.53	**59.69**	38.43
7	**73.42**	65.26	64.00	67.83	54.28	54.61	62.30	66.72
OA	44.83	66.87	78.53	56.73	75.29	84.31	81.24	**88.49**
AA	55.81	64.83	71.72	65.25	66.85	66.29	68.61	**68.74**
Kappa	33.70	57.00	71.38	46.87	66.90	77.85	74.34	**83.69**

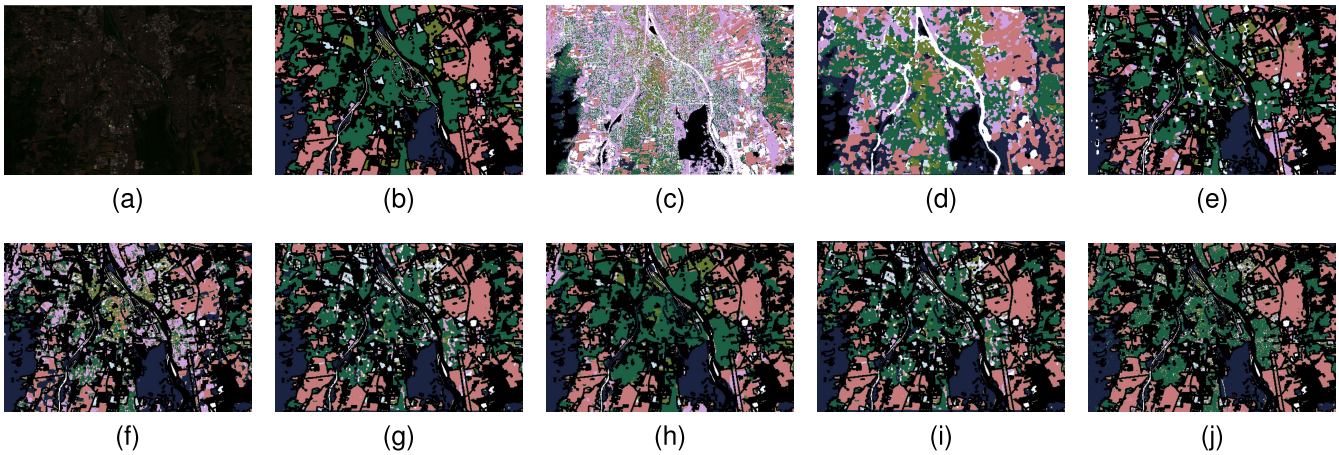


Fig. 11. Classification maps of each algorithm on the Pavia University dataset. (a) Pseudo-color. (b) Ground-truth. (c) EndNet (44.83%). (d) S2ENet (66.87%). (e) Fusion_HCT (78.53%). (f) DSHFNet (56.73%). (g) CMCL (75.29%). (h) CACL (84.31%). (i) NNCNet (81.24%). (j) Proposed method (88.49%).

V. CONCLUSION

In this study, we propose a ternary modality contrastive learning framework aimed at achieving semantic alignment between HSI and LiDAR. This framework segments HSI into spatial and spectral modalities to mitigate the influence of phenomena where similar spectra correspond to different materials. Furthermore, we investigate the formulation of loss functions within a multimodal context to optimize the tradeoff between efficiency and precision. Concurrently, given the distinctive neighborhood similarity characteristic of HSIs, we devise an innovative data augmentation technique to enhance the diversity of the sample set. We conducted extensive experiments on four public datasets, and the results demonstrate that our approach surpasses other SOTA methods. Our method employs a simple CNN architecture for feature extraction. In the future, we aim to utilize more effective feature extraction networks and design more efficient feature fusion modules to facilitate module integration.

APPENDIX A
ROBUSTNESS OF THE MODEL TO NOISE-CONTAMINATED DATA

To simulate potential contaminations in real-world hyperspectral datasets, we experimented by introducing noise to the data. Gaussian noise, with a mean of zero and a variance between 0 and 0.005, was added to 20% of the spectral bands. In addition, salt-and-pepper noise, with intensities ranging from 0 to 0.1, was randomly applied to another 20% of the bands. The impacted spectral bands across each dataset are illustrated in Figs. 12–15. Following noise augmentation, we retrained our model using these modified datasets, with the outcomes detailed in Table VIII.

TABLE VIII
CLASSIFICATION RESULTS OF DIFFERENT DATASETS AFTER ADDING NOISE

	Houston2013	MUUFL	Trento	Augsburg
Without Noise	93.80	83.04	98.58	88.49
Add Noise	91.22	82.15	96.74	83.78

The results indicated that our model's accuracy did not exhibit a drastic reduction. Specifically, the Augsburg dataset, which features complex spatial distributions, irregular class aggregation, and similar features across different classes, faced increased classification difficulty after noise addition, resulting in a relatively larger decline in accuracy compared to the other datasets. Nevertheless, the OA values across all four datasets remained competitive. Notably, even with noise augmentation, the experimental accuracy of the Houston dataset continued to outperform other methodologies tested without noise, underscoring the robustness of our model.

APPENDIX B
GENERALIZATION PERFORMANCE OF THE MODEL ON UNSEEN CLASSES

To validate the efficacy of feature extractors trained via contrastive learning in handling unseen classes, we devised an experimental setup. In the pretraining phase, we selectively

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:37:32 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
CLASSIFICATION PERFORMANCE OF THE MODEL'S GENERALIZATION PERFORMANCE

	Exclude class	Houston	Exclude class	Trento	Exclude class	MUFFL	Exclude class	Augsburg
	1 2 3	91.72	1	93.62	1 2	82.59	1 2	86.67
	4 5 6	91.90	2	94.24	3 4	83.00	3 4	87.04
	7 8 9	91.90	3	94.04	5 6	82.52	5 6	87.19
	10 11 12	90.20	4	94.23	7 8	82.00	6 7	87.06
	13 14 15	90.35	5	92.52	9 10	82.55		
			6	94.68	10 11	82.20		
Average OA		91.21		93.89		82.48		86.99

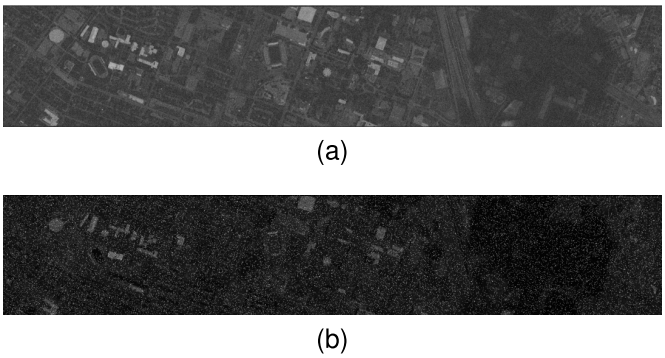


Fig. 12. Visualization of the Houston dataset across different noise-contaminated spectral bands. (a) Gaussian noise 60th band. (b) Salt-and-pepper noise 77th band.

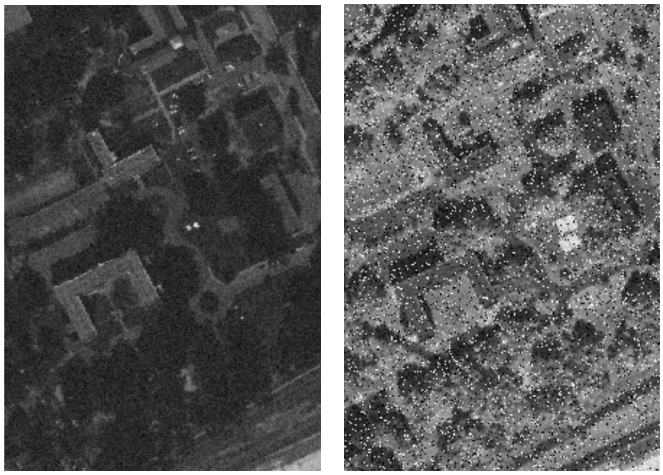


Fig. 13. Visualization of the Houston dataset across different noise-contaminated spectral bands. (a) Gaussian noise 8th band. (b) Salt-and-pepper noise 53th band.

excluded samples from several classes to assess the feature extractor's ability under conditions of incomplete training data. During the transition to downstream classification tasks, we maintained the encoder's parameters static while fine-tuning only the terminal classification layer. The results, detailed in Table IX, list the "exclude class," indicating categories omitted during pretraining.

The findings demonstrate a modest decline in OA, with the MuFFL model showing a decrease of merely 0.56%. This suggests that the pretext task of our self-supervised learning framework, which treats each sample as a unique class to explore the intrinsic attributes of the data, when the diversity of the sample space is sufficiently rich, the model can extract common features of the data, thereby enabling good generalization performance on unseen classes. However, for the Trento dataset, which has a limited number of training samples and classes, thereby lacking sufficient data diversity, the accuracy declined more significantly when a category was excluded from pretraining. This underscores the model's dependency on comprehensive pretraining to achieve optimal performance.

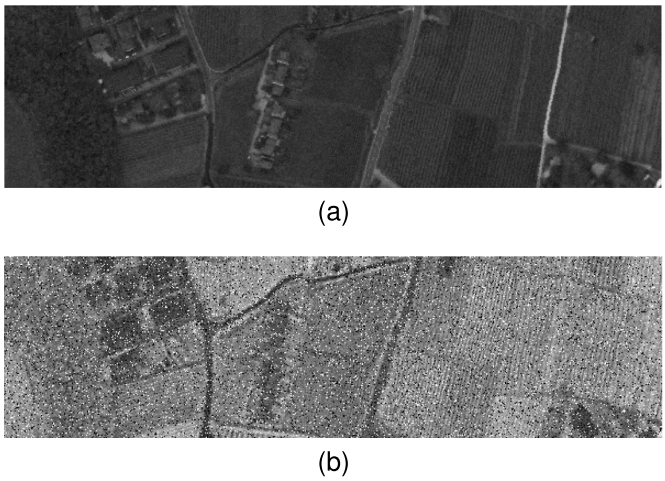


Fig. 14. Visualization of the Houston dataset across different noise-contaminated spectral bands. (a) Gaussian noise 33th band. (b) Salt-and-pepper noise 42th band.

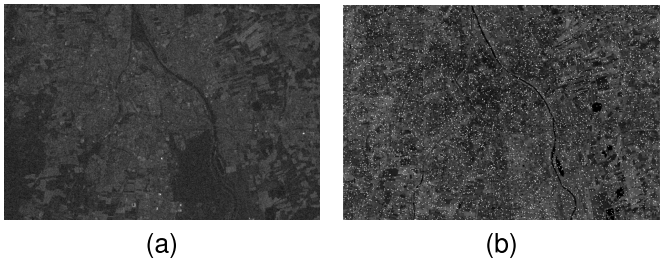


Fig. 15. Visualization of the Houston dataset across different noise-contaminated spectral bands. (a) Gaussian noise 144th band. (b) Salt-and-pepper noise 45th band.

REFERENCES

[1] D. Datta, P. K. Mallick, A. K. Bhoi, M. F. Ijaz, J. Shafi, and J. Choi, "Hyperspectral image classification: Potentials, challenges, and future directions," *Comput. Intell. Neurosci.*, vol. 2022, pp. 1–36, Apr. 2022.

[2] L. W. Kuswidiyanto, H.-H. Noh, and X. Han, "Plant disease diagnosis using deep learning based on aerial hyperspectral images: A review," *Remote Sens.*, vol. 14, no. 23, p. 6031, Nov. 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:37:32 UTC from IEEE Xplore. Restrictions apply.

[3] F. J. Galdames, C. A. Perez, P. A. Estévez, and M. Adams, "Rock lithological instance classification by hyperspectral images using dimensionality reduction and deep learning," *Chemometric Intell. Lab. Syst.*, vol. 224, May 2022, Art. no. 104538.

[4] V. I. Pasha and D. B. Megherbi, "A deep learning approach for hyperspectral image classification with additive noise for remote sensing and airborne surveillance," in *Proc. IEEE 9th Int. Conf. Comput. Intell. Virtual Environments Meas. Syst. Appl. (CIVEMSA)*, Jun. 2022, pp. 1–6.

[5] L. D. Medus, M. Saban, J. V. Francés-Víllora, M. Bataller-Mompeán, and A. Rosado-Muñoz, "Hyperspectral image classification using CNN: Application to industrial food packaging," *Food Control*, vol. 125, Jul. 2021, Art. no. 107962.

[6] M. Ahmad et al., "Hyperspectral image classification—Traditional to deep models: A survey for future prospects," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 968–999, 2022.

[7] R. Roriz, J. Cabral, and T. Gomes, "Automotive LiDAR technology: A survey," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 7, pp. 6282–6297, Jul. 2022.

[8] T. Zhang et al., "SAR ship detection dataset (SSDD): Official release and comprehensive data analysis," *Remote Sens.*, vol. 13, no. 18, p. 3690, Sep. 2021.

[9] W. Yanshuang and Z. Xiaoli, "Object-oriented tree species classification with multi-scale texture features based on airborne hyperspectral images," *J. Beijing Forestry Univ.*, vol. 42, no. 6, pp. 91–101, 2020.

[10] R. Huang and J. Zhu, "Using random forest to integrate LiDAR data and hyperspectral imagery for land cover classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2013, pp. 3978–3981.

[11] X. Xu, J. Li, and A. Plaza, "Fusion of hyperspectral and LiDAR data using morphological component analysis," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, 2016, pp. 3575–3578, doi: 10.1109/IGARSS.2016.7729926.

[12] B. Rasti, P. Ghamisi, and R. Gloaguen, "Hyperspectral and LiDAR fusion using extinction profiles and total variation component analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3997–4007, Jul. 2017.

[13] D. L. Donoho et al., "High-dimensional data analysis: The curses and blessings of dimensionality," *AMS Math Challenges Lect.*, vol. 1, p. 32, Aug. 2000.

[14] F. Murtagh and M. M. Farid, "Pattern classification, by Richard O. Duda, Peter E. Hart, and David G. Stork," *J. Classification*, vol. 18, no. 2, pp. 273–275, 2001.

[15] G. Cheng, X. Xie, J. Han, L. Guo, and G.-S. Xia, "Remote sensing image scene classification meets deep learning: Challenges, methods, benchmarks, and opportunities," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3735–3756, 2020.

[16] X. Wu, D. Hong, and J. Chanussot, "Convolutional neural networks for multimodal remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5517010, doi: 10.1109/TGRS.2021.3124913.

[17] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4939–4950, Jul. 2020.

[18] H. Li, P. Ghamisi, U. Soergel, and X. X. Zhu, "Hyperspectral and LiDAR fusion using deep three-stream convolutional neural networks," *Remote Sens.*, vol. 10, no. 10, p. 1649, Oct. 2018.

[19] S. Fang, K. Li, and Z. Li, "S^{2}ENet: Spatial–spectral cross-modal enhancement network for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[20] Y. Feng, L. Song, L. Wang, and X. Wang, "DSHFNet: Dynamic scale hierarchical fusion network based on multiattention for hyperspectral image and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522514, doi: 10.1109/TGRS.2023.3311535.

[21] X. Zhao et al., "Joint classification of hyperspectral and LiDAR data using hierarchical random walk and deep CNN architecture," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7355–7370, Oct. 2020, doi: 10.1109/TGRS.2020.2982064.

[22] X. Wang, Y. Feng, R. Song, Z. Mu, and C. Song, "Multi-attentive hierarchical dense fusion net for fusion classification of hyperspectral and LiDAR data," *Inf. Fusion*, vol. 82, pp. 1–18, Jun. 2022.

[23] Y. Yang, D. Zhu, T. Qu, Q. Wang, F. Ren, and C. Cheng, "Single-stream CNN with learnable architecture for multisource remote sensing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5409218, doi: 10.1109/TGRS.2022.3169163.

[24] S. Jia et al., "Multiple feature-based superpixel-level decision fusion for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1437–1452, Feb. 2020.

[25] D. Xiu, Z. Pan, Y. Wu, and Y. Hu, "MAGE: Multisource attention network with discriminative graph and informative entities for classification of hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5539714, doi: 10.1109/TGRS.2022.3210398.

[26] Y. Liu et al., "Graph self-supervised learning: A survey," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 6, pp. 5879–5900, Jun. 2023.

[27] Y. Wang, C. M. Albrecht, N. A. A. Braham, L. Mou, and X. X. Zhu, "Self-supervised learning in remote sensing: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 4, pp. 213–247, Dec. 2022.

[28] J. Lee, Y. Jeong, and M. Cho, "Self-supervised learning of image scale and orientation," 2022, *arXiv:2206.07259*.

[29] C. Tao et al., "Siamese image modeling for self-supervised vision representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 2132–2141.

[30] M. C. Schiappa, Y. S. Rawat, and M. Shah, "Self-supervised learning for videos: A survey," *ACM Comput. Surv.*, vol. 55, no. 13s, pp. 1–37, Dec. 2023.

[31] J. Yue, L. Fang, H. Rahmani, and P. Ghamisi, "Self-supervised learning with adaptive distillation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501813, doi: 10.1109/TGRS.2021.3057768.

[32] X. Yang, W. Cao, Y. Lu, and Y. Zhou, "Self-supervised learning with prediction of image scale and spectral order for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5545715, doi: 10.1109/TGRS.2022.3225663.

[33] S. Hou, H. Shi, X. Cao, X. Zhang, and L. Jiao, "Hyperspectral imagery classification based on contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2021.

[34] S. Zhang, Z. Chen, D. Wang, and Z. J. Wang, "Cross-domain few-shot contrastive learning for hyperspectral images classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[35] X. Huang, M. Dong, J. Li, and X. Guo, "A 3-D-swin transformer-based hierarchical contrastive learning method for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5411415.

[36] P. Guan and E. Y. Lam, "Cross-domain contrastive learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5528913, doi: 10.1109/TGRS.2022.3176637.

[37] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, vol. 139, 2021, pp. 8748–8763.

[38] J. Li, R. Selvaraju, A. Gotmare, S. Joty, C. Xiong, and S. C. H. Hoi, "Align before fuse: Vision and language representation learning with momentum distillation," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 9694–9705.

[39] J. Lu, D. Batra, D. Parikh, and S. Lee, "ViLBERT: Pretraining task-agnostic visiolinguistic representations for vision-and-language tasks," in *Proc. 33rd Int. Conf. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 13–23.

[40] H. Bao et al., "VLMo: Unified vision-language pre-training with mixture-of-modality-experts," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 32897–32912.

[41] R. Hang, X. Qian, and Q. Liu, "Cross-modality contrastive learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, doi: 10.1109/TGRS.2022.3188529.

[42] M. Wang, F. Gao, J. Dong, H.-C. Li, and Q. Du, "Nearest neighbor-based contrastive learning for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501816.

[43] X. Chen and K. He, "Exploring simple Siamese representation learning," in *Proc. IEEE Comput. Soc. Conf. Comput. Vision Pattern Recognit.*, Jun. 2021, pp. 15750–15758.

[44] S. Jia, X. Zhou, S. Jiang, and R. He, "Collaborative contrastive learning for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507714, doi: 10.1109/TGRS.2023.3263511.

[45] W. Song, S. Li, L. Fang, and T. Lu, "Hyperspectral image classification with deep feature fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3173–3184, Jun. 2018.

[46] S. Xia, X. Zhang, H. Meng, J. Fan, and L. Jiao, "Two-stream networks for contrastive learning in hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 1903–1920, 2024.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:37:32 UTC from IEEE Xplore. Restrictions apply.



Shuxiang Xia received the B.S. degree in computer science and technology from the Xi'an University of Technology, Xi'an, China, in 2022. He is currently pursuing the master's degree with Xidian University, Xi'an.

His main research interests include deep learning, computer vision, and hyperspectral image (HSI) classification.



Xiaohua Zhang (Member, IEEE) received the B.S. and M.S. degrees in applied mathematics from Northwest University, Xi'an, Shaanxi, China, in 1997 and 2000, respectively, and the Ph.D. degree in circuits and systems from Xidian University, Xi'an, in 2004.

In 2009, he was a Visiting Scholar with the Georgia Institute of Technology, Atlanta, GA, USA. He is currently an Associate Professor with the School of Artificial Intelligence, Xidian University, and a member of the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education of China. His research interests include image processing, deep learning, computer vision, and remote sensing image processing.

Dr. Zhang is a Senior Member of the Chinese Institute of Electronics and the China Computer Federation.



Hongyun Meng received the Ph.D. degree in operating research and control from the School of Mathematics and Statistics, Xidian University, Xi'an, China, in 2004.

She is currently an Associate Professor with the School of Mathematics and Statistics, Xidian University. Her research interests include intelligent information processing, natural computing, and multiobjective optimization.



Licheng Jiao (Fellow, IEEE) received the B.S. degree in electrical and computer science from Shanghai Jiao Tong University, Shanghai, China, in 1982, and the M.S. and Ph.D. degrees in theoretical electrician from Xi'an Jiaotong University, Xi'an, China, in 1984 and 1990, respectively.

Since 1992, he has been a Professor with the School of Electronic Engineering, Xidian University, Xi'an, where he is currently the Director of the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education of China. His research interests include image processing, natural computation, machine learning, and intelligent information processing.

Dr. Jiao is a Foreign Member of the Academia Europaea and the Russian Academy of Natural Sciences. He is a fellow of the Institution of Engineering and Technology, the Chinese Association for Artificial Intelligence, the Chinese Institute of Electronics, the China Computer Federation, and the Chinese Association of Automation. He is a Councilor of the Chinese Institute of Electronics, a Committee Member of the Chinese Committee of Neural Networks, and an Expert of the Academic Degrees Committee of the State Council. He is the Chairperson of the Awards and Recognition Committee and the Vice Board Chairperson of the Chinese Association of Artificial Intelligence.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:37:32 UTC from IEEE Xplore. Restrictions apply.